

Data-Driven Insights into Competitive Electricity Supply Participation in Massachusetts

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I. Executive Summary

199K kWh/mo Average Large C&I	84.4% Competitive Adoption Large C&I	22.3% Competitive Adoption Residential	37% Municipal Aggregation Residential	97% Accuracy ML Model (Classification)
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This analysis examines electricity consumption and competitive supply participation across six customer segments in Massachusetts from 2020 to 2025, using monthly data from the Department of Energy Resources (DOER)¹. The analysis finds that **customer class is the single strongest predictor** of how much electricity a customer consumes and how likely they are to leave basic utility service for a competitive supplier or municipal aggregation program.

Consumption is highly concentrated: Large commercial & industrial (C&I) customers average **199,000 kilowatts per hour (kWh) per month**, roughly 350 times more than residential customers, yet residential accounts collectively hold the largest share of total demand at 34.4%. Competitive adoption mirrors this pattern. **Large C&I leads at 84.4%** direct competitive adoption, while residential sits at 22.3%, though an additional 37% participate through municipal aggregation, meaning most residential customers have, in fact, left basic service through municipal-level community programs.

A machine learning classification model confirmed these dynamics, predicting the dominant supply type with **97% accuracy** using only lagged historical adoption rates. Retail competition is working well for high-volume users; extending its benefits to smaller segments requires expanded municipal aggregation, simplified competitive products, and digital acquisition channels. Competitive suppliers serving Large C&I should shift focus toward retention, demand flexibility services, and sustainability reporting to deepen relationships and raise switching costs.

II. Problem Statement

Electricity markets in many U.S. states have introduced competitive retail supply, allowing customers to choose between their incumbent utility provider and alternative competitive suppliers. This project analyzes how electricity consumption and customer participation vary across different customer segments in Massachusetts.

The objective of this study is to examine how customer class influences electricity consumption patterns and participation in competitive electricity supply markets. By analyzing electric customer choice data² from January 2020 to August 2025 from the DOER, the project will explore whether certain customer segments, such as residential or commercial customers, are more likely to adopt competitive electricity suppliers and how electricity usage differs across these segments. Understanding these patterns provides insight into customer behavior in partially deregulated electricity markets and can help evaluate how competition affects energy demand across sectors.

¹Mass.gov, Electric & Gas Customer Choice Data, accessed 3/9/26, <https://www.mass.gov/info-details/electric-gas-customer-choice-data>.

²Electric Customer Choice Data 2020, Q3 2025, <https://www.mass.gov/doc/electric-customer-choice-data-q2-2025-spreadsheet/download>.

III. Dataset Description

The original DOER dataset uses a grouped multi-level layout with merged header rows (utility names span multiple columns, metric types are nested beneath them, and supplier categories form a third header row). Year and month identifiers are embedded in a hierarchical row structure rather than appearing as flat columns. This layout is designed for human readability but is not compatible with Tableau, Power BI, or direct analytical consumption by pandas/R. These data are reported monthly by investor-owned electric utilities and include the two key variables—number of customers in each rate class and total electricity consumption (kWh) for the month. The variables are reported separately for incumbent utility generation service and competitive electricity suppliers. Customer classes include residential, farm, streetlights, small C&I ($\leq 3,000$ kWh/month), medium C&I (3,000–120,000 kWh/month), and large C&I ($>120,000$ kWh/month). Each of these variables is grouped as a time series of total annual consumption from 2020 to 2025.

IV. Analysis Methodology

The analysis methodology applied to this electricity consumption dataset (2020-2025) involved several key steps, focusing on understanding consumption patterns, customer demographics, and competitive supplier adoption. The initially sourced dataset was nested and would be almost impossible to analyze. To address this, the data was flattened so that each row represents a unique combination of year, month, customer class, and supply type, with all values consolidated into a single column. A consistent color palette was established for various customer class categories to maintain visual clarity across all charts. The analysis addressed four core questions. First, the relationship between customer segments, consumption volume, and supplier type was examined to understand how electricity sourcing varies across rate classes. Second, electricity usage was compared across segments, including each segment's share of total demand. Third, a trend analysis tracked changes in customer participation in competitive versus incumbent electricity service over time. Fourth, a ratio analysis computed average electricity consumption per customer, calculated as “kWh Used” divided by “Number of Customers” to normalize comparisons across segments of different sizes. “Total” supply type rows were excluded from this calculation to avoid double-counting. This comprehensive approach enabled a multifaceted understanding of electricity consumption and competitive market dynamics across customer segments over the specified period.

Two machine learning models are applied for the prediction of customer consumption patterns and the supply adoption by customer class. Due to the underrepresentation of the Basic service class in the data, the machine learning model was normalized using a logarithmic transformation to mitigate skewness. The models are based on the aggregated monthly dataset spanning January 2020 through August 2025, covering seven customer classes across three supply types. The classification model addressed two core business questions: which customer segments are most likely to adopt competitive electricity suppliers, and whether a segment's movement away from basic service can be predicted using historical participation patterns. Data are classified into 3 groups based on the target variables - supply type (Basic Service, Competitive Supply, or Municipal). The target labels are derived by comparing customer counts across the three supply types for each unique combination of month, year, and customer class. The feature set will include customer class, year, month, customer counts per supply type, and historical adoption rates, specifically the Basic Service rate, Competitive Adoption rate, and Municipal Aggregation

rate from prior periods. The regression model will address two business questions: how much electricity a given customer segment is expected to consume given its rate class and supplier mix, and whether switching to competitive supply is associated with higher or lower consumption levels when controlling for segment size. The regression model will be developed to predict average electricity consumption per customer, calculated as “total kWh” divided by “total customer count” for a given segment, supplier type, and time period. The feature set will include customer class, supply type distribution, adoption rates, and month and year to account for seasonal and longitudinal variation in consumption.

V. Data Limitations

Data for Municipal Aggregation is largely null for most of 2020 and parts of 2021. This reflects a gradual rollout of community choice programs. Therefore, time-series analysis focusing on this supplier type should ideally commence from 2022 onwards, or null values for earlier periods should be treated as zero (indicating no program was active). The Farms customer class data is only available for Eversource. National Grid and utilities show null values across all Farm rows. Any cross-utility comparisons involving this customer class should either exclude it or explicitly footnote this data inconsistency. The dataset includes Residential Low-Income as a distinct sub-segment, separate from general Residential. The current analysis or proposal did not explicitly address this class, so its treatment (whether as an additional category or merged with general Residential) should be clarified for a complete understanding of the residential segment. The analysis proceeds with the assumption that the detailed and aggregated files, after flattening and exclusion of subtotal rows (to prevent double-counting), are clean and ready for analytical tools.

VI. Key Findings

The analysis of electricity consumption and supplier choice patterns reveals several important insights. As shown in Figure 1, residential customers account for the largest share of total electricity consumption at 34.4%, followed closely by Large C&I customers at 31.7%. However, this picture changes significantly when consumption is normalized per customer (Figure 2). A single Large C&I customer consumes an average of 199,321 kWh, approximately 350 times more than a typical residential customer at 557 kWh. This contrast highlights that residential dominance in total consumption is driven entirely by the sheer volume of residential accounts and not by individual customer usage. Together, these two classes represent two-thirds of all electricity consumed in the service territory (Figure 3).

Seasonal patterns differ significantly across customer classes. Residential consumption exhibits the strongest seasonal variation, with pronounced peaks driven by winter heating and summer cooling, whereas Large C&I usage remains relatively stable with only mild fluctuations (Figure 4). When aggregated across all classes, system-wide electricity demand reaches its highest levels in July to August due to cooling needs and shows a secondary peak in January to February associated with heating; demand is lowest during the shoulder months of April to May and October to November (Figure 5).

With respect to the supplier market participation based on customer class, Farms are the most basic service-dependent class, with roughly 59% still on basic service, as shown in Figure 6. The C&I customer segments are concentrated on competitive supply contracts. A significant share of

residential customers (37%) participate in Municipal Aggregation programs, which might have been incentivized by community-level programs.

Figure 7 explains the competitive supply adoption dynamics, with Large C&I customers displaying the highest engagement at 84.4% competitive adoption rate, indicating the vast majority of these customers have moved away from basic utility service. However, residential customers have the lowest switching rate at 22.3 percent.

The examination of trends in Figure 8 from 2020 to 2025 shows a significant disruption in the competitive supplier participation across all customer segments in 2021. Residential competitive adoption saw a sharp decline from approximately 58% to below 20%, partially recovering to 22% in 2025. On the other hand, Large C&I adoption recovered quickly and has remained consistently stable between 80%-88%. There seems to be a structural market shift in 2020 where residential customers switch from basic utility services and competitive suppliers to municipal aggregation, which grew to become a major segment by 2025 (Figure 9). The competitive electricity market seems to be structurally oriented toward commercial and industrial customers. The steady yet consistent decline of the basic service suggests a gradual shift away from incumbent utility providers to municipal utility aggregation rather than the competitive suppliers.

Figure 10 demonstrates a strong positive relationship between average monthly consumption and competitive supplier adoption rates. Larger users, particularly Large C&I accounts, exhibit a significantly higher propensity to engage with competitive suppliers, confirming that consumption volume is a key driver of competitive supplier adoption across customer classes.

As reported in Figure 11, the classification model achieved strong overall performance, correctly identifying the dominant supply type in 97% of test cases. It proved especially reliable for Competitive and Municipal categories, with precision, recall, and F1-scores approaching 1.00. The most influential predictors were the two-month lagged municipal aggregation adoption rate, the one-month lagged competitive adoption rate, and the one-month lagged basic service rate, underscoring the predictive power of recent historical participation trends.

By contrast, the regression model performed poorly when predicting average kWh per customer. The R^2 score of $-6,395.09$ indicates that its predictions were substantially worse than simply using the sample mean, while the root mean squared error reached 32,178.80. Feature importance analysis revealed that the Large C&I customer class exerted an overwhelming influence (importance weight of 0.996), suggesting that extreme consumption values in this segment, combined with the high variance across classes, limited the model's ability to generalize. The regression model after class normalization by taking the log of each class improved with a negative R^2 score of -2500.79 , MSE of 405019637.66 and RSME of 20125.10. While this is an improvement from the previous model, it still performed extremely poor with those summary statistics, thus unreliable. These results point to the need for additional data exploration or model refinement focused on the Large C&I segment before reliable consumption forecasts can be obtained. Taken together, the machine-learning analysis reinforces the descriptive findings presented earlier. Historical adoption rates alone provide a robust basis for anticipating shifts in supplier choice, yet predicting per-customer consumption levels remains challenging given the pronounced differences in scale across customer classes.

VI. Conclusion

This study demonstrates that the customer class is a central driver of electricity consumption patterns and competitive supply participation. Large C&I customers, though few in number, consume nearly as much electricity as the entire residential sector while maintaining consistently high direct competitive adoption. Residential customers dominate total usage but rely overwhelmingly on municipal aggregation rather than individual switching. The analysis confirms a strong positive relationship between average monthly consumption and the propensity to leave basic service, alongside distinct seasonal demand profiles across classes.

Machine learning results reinforce these patterns. The classification model achieved 97% accuracy in predicting dominant supply type using only lagged historical adoption rates, confirming the predictive power of recent participation trends. The regression model, however, performed poorly in forecasting average kWh per customer due to extreme variance from large industrial accounts.

Overall, retail competition functions effectively for high-volume users but requires community-level mechanisms to engage smaller residential and Small C&I customers. These segmented dynamics provide practical guidance for utilities, regulators, and suppliers in program design, demand forecasting, and evaluating partial market deregulation. Future research should develop class-specific models and incorporate price or demographic variables.

VII. Recommendations

Competitive suppliers should continue targeting Large C&I accounts, where adoption rates remain consistently high. Future forecasting efforts should develop class-specific models to address the extreme variance in large industrial consumption and incorporate price differentials and demographic variables to strengthen predictive accuracy.

- **Defend and Deepen Large C&I:** Shift performance metrics from acquisition to retention, using lagged participation data from the classification model to flag disengaging accounts early, given that this segment drives 31.7% of total consumption and losing even a few accounts creates significant procurement and revenue exposure.
- **Prioritize Medium C&I as a growth market:** This class shows the most dynamic growth opportunity in the competitive market, with 60.34% adoption rate and a large enough customer base for consumption. Target the 39.7% still on basic service by using DOER data to identify the lowest-penetration utility territories first, and offer standardized fixed-price products that are easier to hedge given the segment's relatively stable seasonal demand profile.
- **Develop targeted residential re-engagement strategies:** Maximize the 34.4% total consumption opportunity of the residential class by offering residential-specific products to address the low per-customer consumption profile, such as community-based pricing, bundled services, or simplified fixed-rate contracts that raise switching costs and make competitive market participation viable.
- **Investigate and address the causes of the 2021 market disruption:** A review of the decline in the competitive market adoption across all classes in 2021, leading to a disproportionate and long-lasting consequential impact on the residential class, will reveal where intervention for resilience and recovery is needed.

Appendix

Figure 1. Residential customers have the largest share of the total electricity consumption at 34.4%, slightly higher than Large C&I at 31.7%. However, the story changes drastically when normalized per customer, with the Large C&I customer consuming roughly 360 times more electricity than the residential customer.

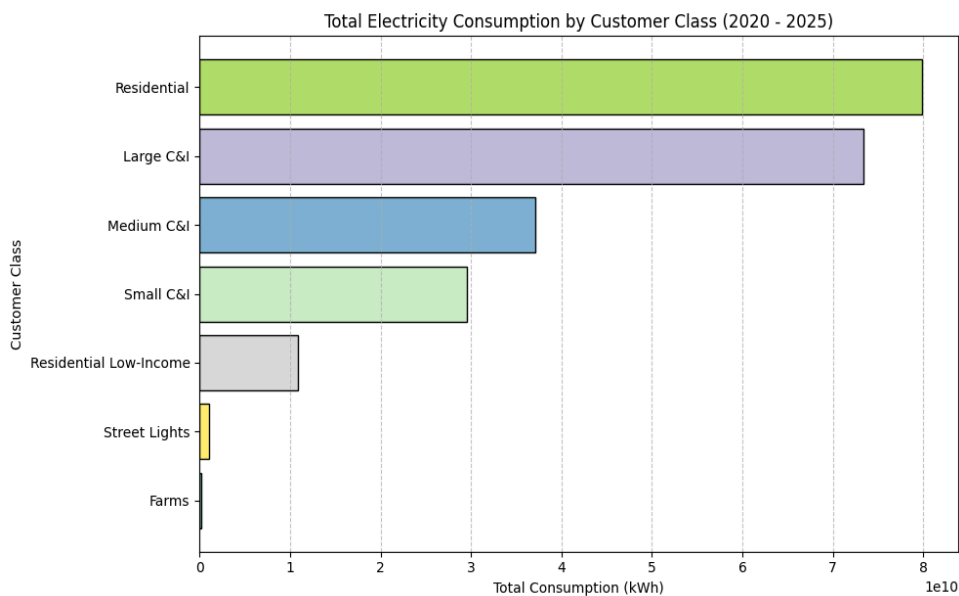


Figure 2. Commercial and industrial customers cluster at the top while residential customers are at the bottom, with Large C&I customers having the highest consumption at an average of 199,000 kWh per month per customer, about 350x more, compared to Residential. This huge difference highlights the extreme consumption concentration in a small number of industrial accounts.

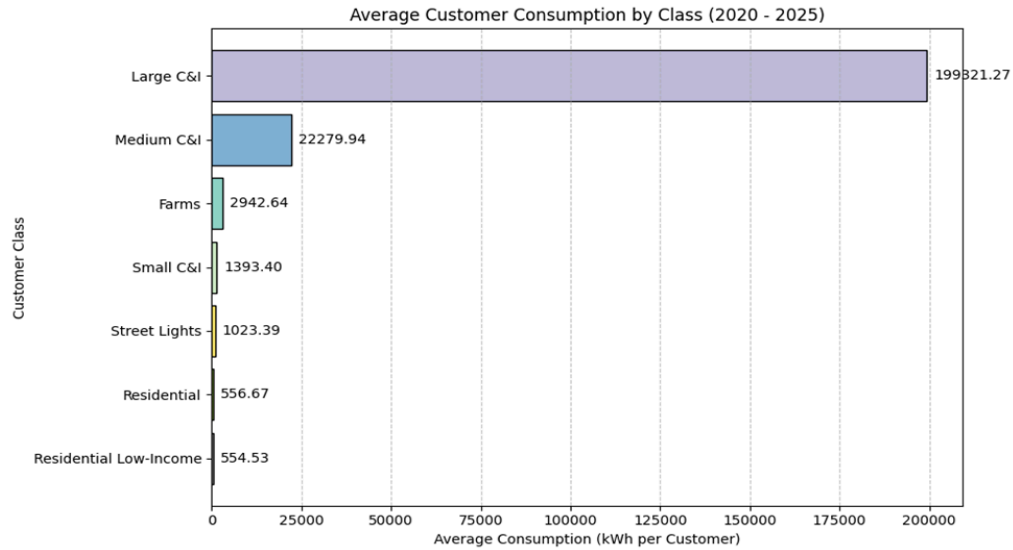


Figure 3. Two classes (Residential and Large C&I) together account for 66% of all electricity consumed, despite having vastly different customer profiles.

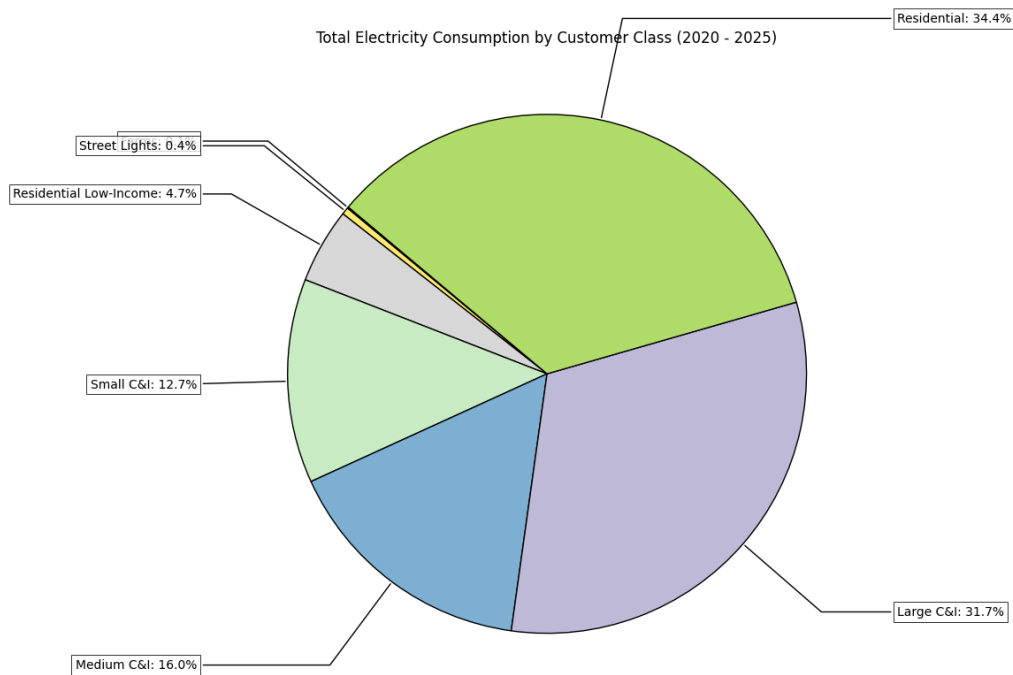


Figure 4. Residential consumption shows the strongest seasonal swing (winter heating and summer cooling peaks). Large C&I consumption is more stable but still shows mild seasonal variation. The time series confirms that seasonal patterns differ significantly by class.

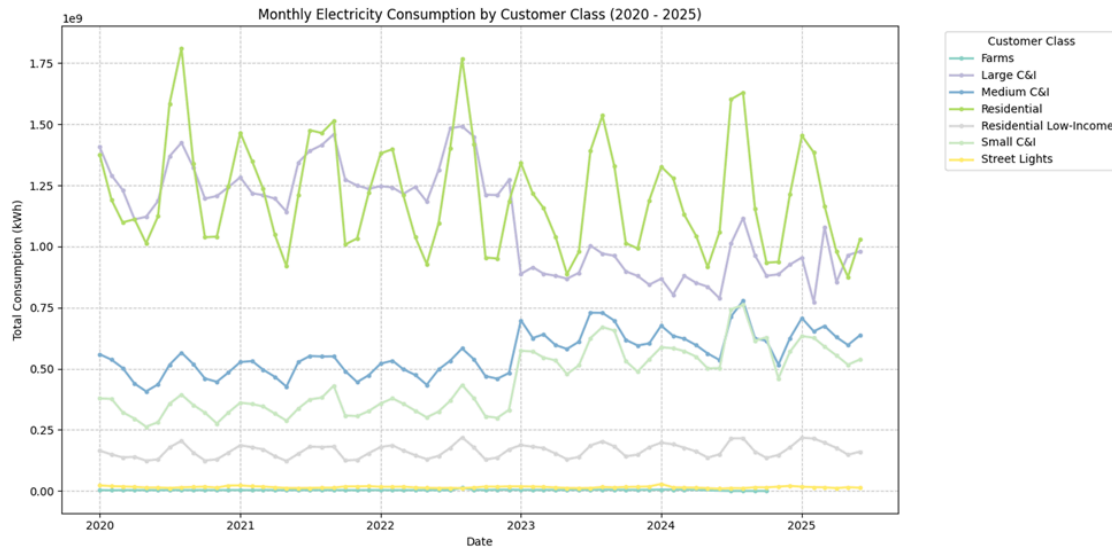


Figure 5. The highest consumption is in summer (July-August), with August being the single highest month at roughly 4.35 billion kWh.

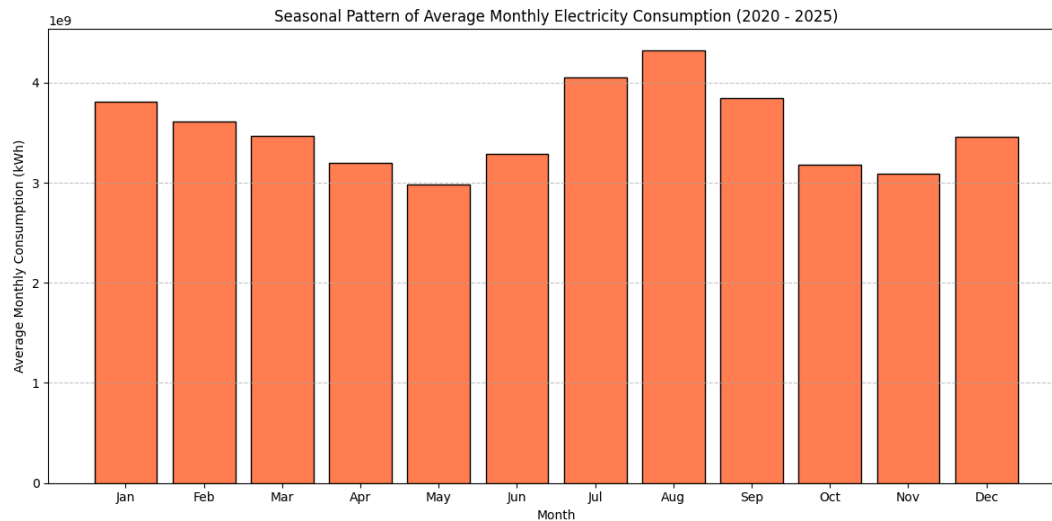


Figure 6. Farms are the most basic service-dependent class, with roughly 59% still on basic service; farms have the lowest overall departure from the incumbent utility. Residential customers are not simply staying on Basic Service; a significant share (37%) participates in Municipal Aggregation programs, which might have been incentivized by community-level programs rather than individual switching. For Large C&I, competitive supply dominates overwhelmingly.

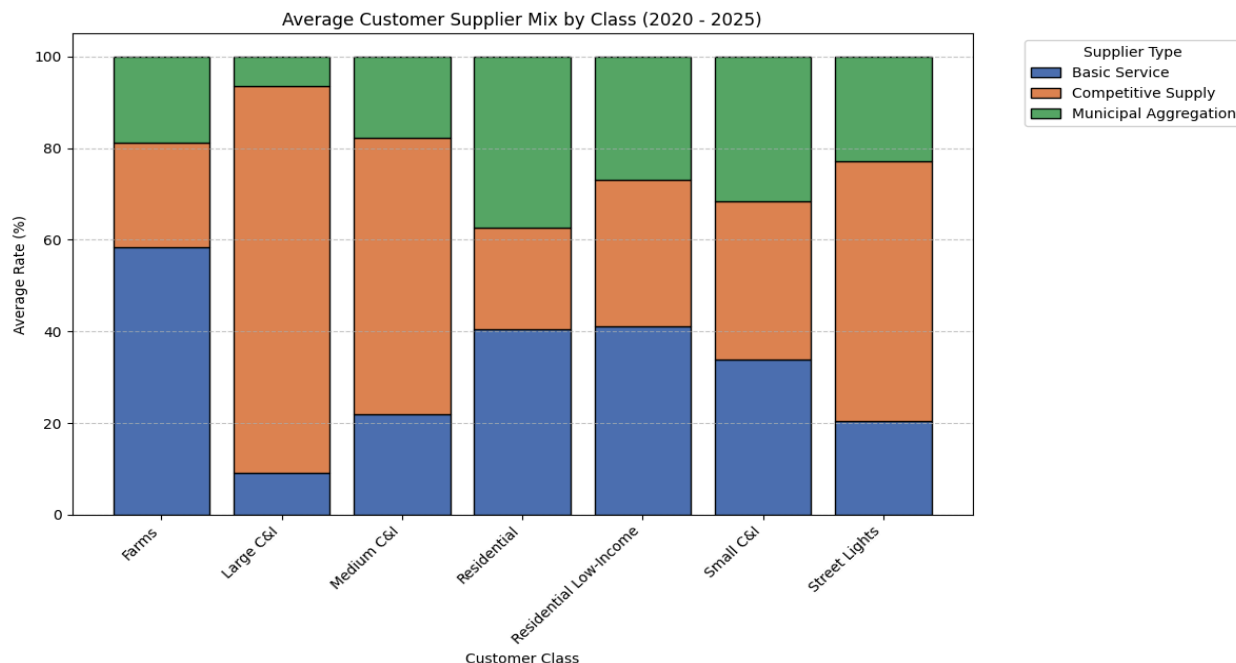


Figure 7. Large C&I customers lead with an 84.4% competitive supplier adoption rate, meaning the vast majority have switched away from basic utility service. Medium C&I follows at 60.3%, and Street Lights at 56.6%. Residential customers have the lowest rate at 22.3%, though a large share also participates in the Municipal Aggregation.

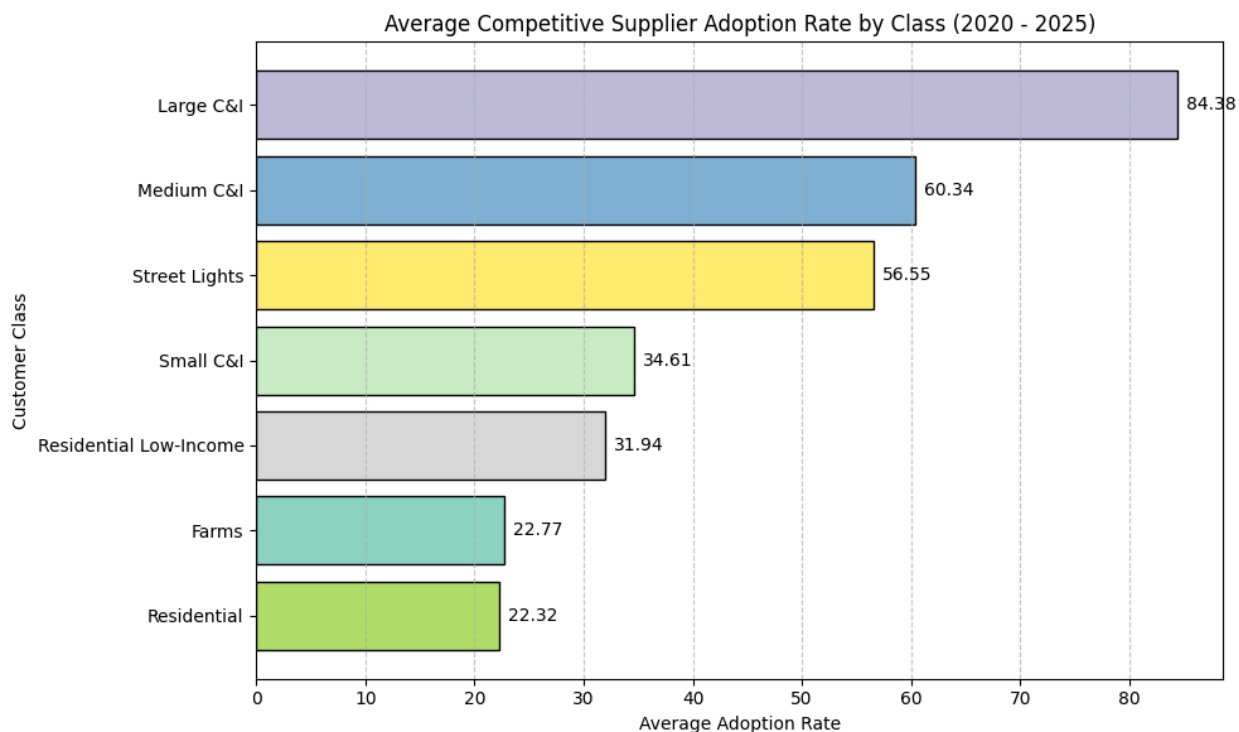


Figure 8. All four classes experienced a sharp decline in competitive adoption around early 2021. Residential drops most dramatically from around 58% to below 20% almost instantaneously, while large C&I adoption has remained consistently high between 80 - 88% throughout the period.

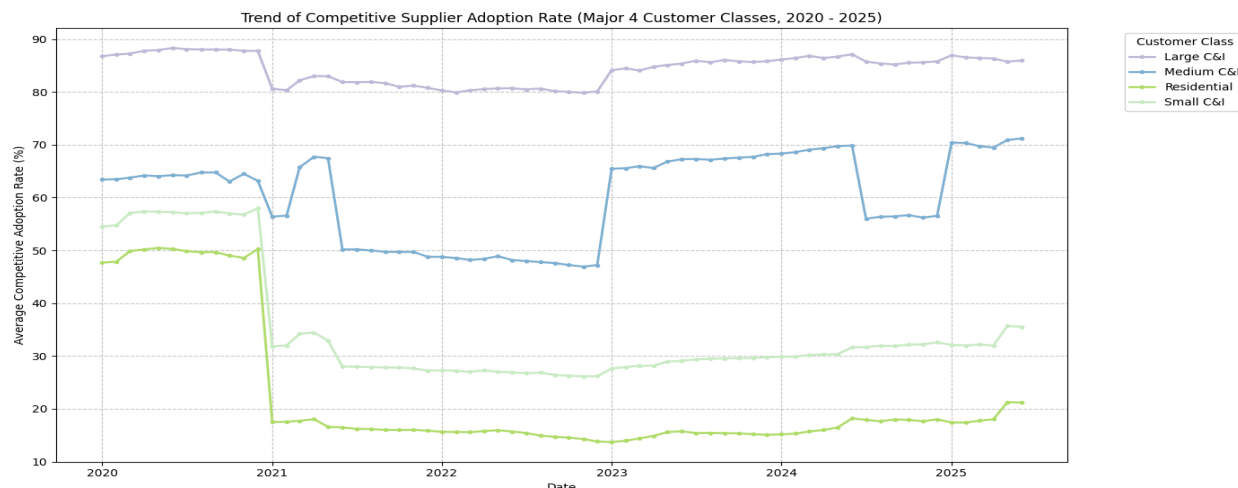


Figure 9. This stacked area chart shows a market shift in 2021, with the Municipal Aggregation segment growing its customer base substantially from near-zero in early 2020 to a major share by 2025. Basic Service customers steadily declined, while Competitive Supply saw a sharp decline in its customer participation.

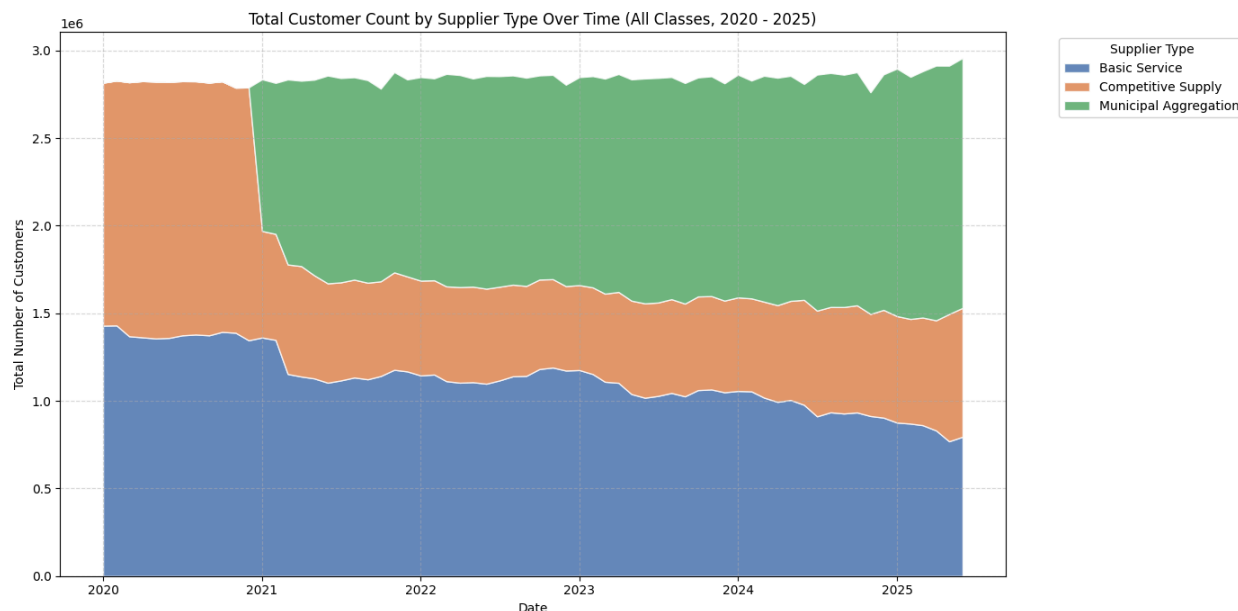


Figure 10. The scatter/bubble plot reveals a clear positive relationship between average customer consumption (x-axis, log scale) and competitive supplier adoption (y-axis). Large C&I customers, who consume 199,000 kWh/month on average, have an 84% adoption rate. Residential customers, at 557 kWh/month, sit at just 22%. The pattern suggests that larger energy

users have stronger economic incentives to seek a competitive supplier, consistent with the hypothesis across most classes, with notable exceptions to the farm class and streetlight.

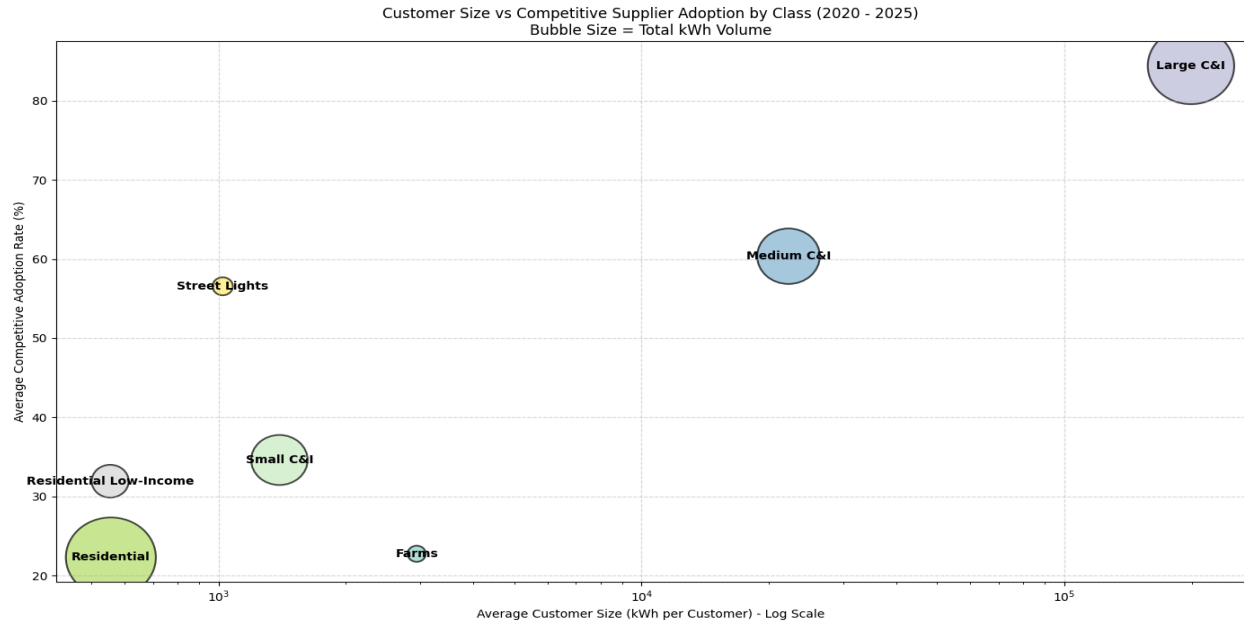


Figure 11. The classification predictive model shows good accuracy for Municipal and Competitive, with an average F1 score of 0.97, but its overall reliability is questionable due to the absence of the ‘Basic’ class in the test data.

Classification Report:				
	precision	recall	f1-score	support
Basic	0.00	0.00	0.00	0
Competitive	1.00	0.95	0.98	65
Municipal	1.00	1.00	1.00	23
accuracy			0.97	88
macro avg	0.67	0.65	0.66	88
weighted avg	1.00	0.97	0.98	88

Top 10 most important features for classification:

MuniAgg_Adoption_Rate_lag2	0.374806
Competitive_Adoption_Rate_lag1	0.354018
Basic_Service_Rate_lag1	0.074239
Basic_Service_Rate_lag2	0.063126
MuniAgg_Adoption_Rate_lag1	0.062613
Month	0.016021
Customer_Class_Small C&I	0.015938
Customer_Class_Residential Low-Income	0.014329
Year	0.013266
Customer_Class_Residential	0.008812

dtype: float32

Figure 12. The regression model performs extremely poorly with a negative R^2 score of -6395.0925 and high MSE and RMSE, largely due to an over-reliance on a single customer class ‘Large C & I’ due to their predictive weight in the dataset.

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Regression R2 Score: -6395.0925
Mean Squared Error: 1035474874.54
Root Mean Squared Error: 32178.80

Top 10 most important features for regression:
Customer_Class_Large C&I      0.995842
Year                          0.001754
Basic_Service_Rate            0.000866
Basic_Service_Rate_lag1       0.000722
Customer_Class_Medium C&I     0.000505
Competitive_Adoption_Rate     0.000153
MuniAgg_Adoption_Rate_lag1    0.000032
MuniAgg_Adoption_Rate         0.000029
MuniAgg_Adoption_Rate_lag2    0.000027
Basic_Service_Rate_lag2       0.000027
dtype: float32
    
```

Figure 13. The regression model after normalization improved with a negative R² score of -2500.7923, MSE of 405019637.66 and RSME of 20125.10. While this is an improvement from the previous model, it still performed extremely poorly with those summary statistics, thus unreliable.

